higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

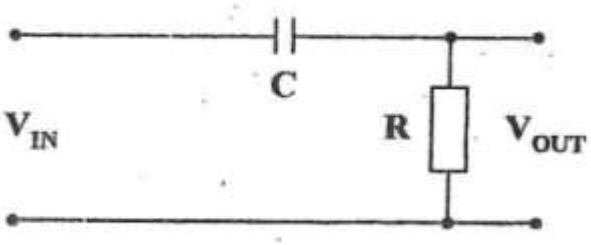
3 APRIL 2019

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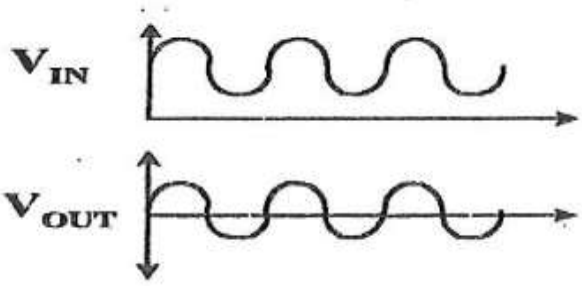
QUESTION 1: ALTERNATING CURRENT THEORY

1.1



✓✓

Coupling circuit



✓✓

Input and output waveforms (4)

1.2 As soon as a perfect square waveform is sent through a faulty circuit, the existence of an error✓ as well as the type of error will be indicated while sine waves often only indicate the existence of an error.✓ (2)

1.3 $Z_1 = 150 + j399\ \Omega = 426{,}264 \angle 69{,}4°\ \Omega$

$Z_2 = 250 - j159\ \Omega = 296{,}28 - \angle 32{,}5°\ \Omega$

$$Z_p = \frac{Z_1 \times Z_2}{Z_1 + Z_2}$$

$$= \frac{426{,}264 \angle 69{,}4° \times 296{,}28\ -\angle 32{,}5°}{150 + j399 + 250 - j159}\ ✓$$

$$= \frac{126\ 293{,}498 \angle 36{,}9°}{400 + j240}\ ✓$$

$$= \frac{126\ 293{,}498 \angle 36{,}9°}{466{,}476\ \angle 30{,}964°}\ ✓$$

$$= 270{,}74\ -\angle 5{,}936°\ \Omega\ ✓$$

$$= 269{,}288 - j27{,}999\ \Omega$$

$$Z_T = Z_P + Z_3$$

$$= 269{,}288 - j27{,}999 + 100 + j257$$

$$= 369{,}288 + j229{,}001 = 434{,}528 \angle 31{,}80\ \Omega \checkmark$$

$$I_T = \frac{V_T}{Z_T}$$

$$= \frac{250 \angle 0°}{434{,}528 \angle 31{,}80\ \Omega} \checkmark$$

$$= -0{,}575 - \angle 31{,}80\ A \checkmark$$

$$V_P = I_T Z_P$$

$$= 0{,}575 - \angle 31{,}80° \times 270{,}74 - \angle 5{,}936°\ \Omega \checkmark$$

$$= 155{,}76 - \angle 37{,}736\ V \checkmark \longrightarrow$$ (9)

[15]

QUESTION 2: POWER SUPPLIES

2.1 $$V'_{DC} = \frac{R_L}{R_L + R} \times V_{DC}$$

$$= \frac{5\,000}{5\,000 + 250} \times 150 \checkmark$$

$$= 142{,}86\ V \checkmark$$

$$V'_{r(rms)} = \frac{X_C}{\sqrt{R^2 + X_C^2}} \times V_{r(rms)}$$

$$Xc = \frac{1}{2\pi fC}$$

$$= \frac{1}{2 \times \pi \times 50 \times 6 \times 10^{-6}}$$

$$= 241{,}144\ \Omega \checkmark$$

$$V'_{r(rms)} = \frac{X_C}{\sqrt{R^2 + X_C^2}} \times V_{r(rms)}$$

$$= \frac{241{,}144}{\sqrt{250^2 + 241{,}144^2}} \times 10 \checkmark$$

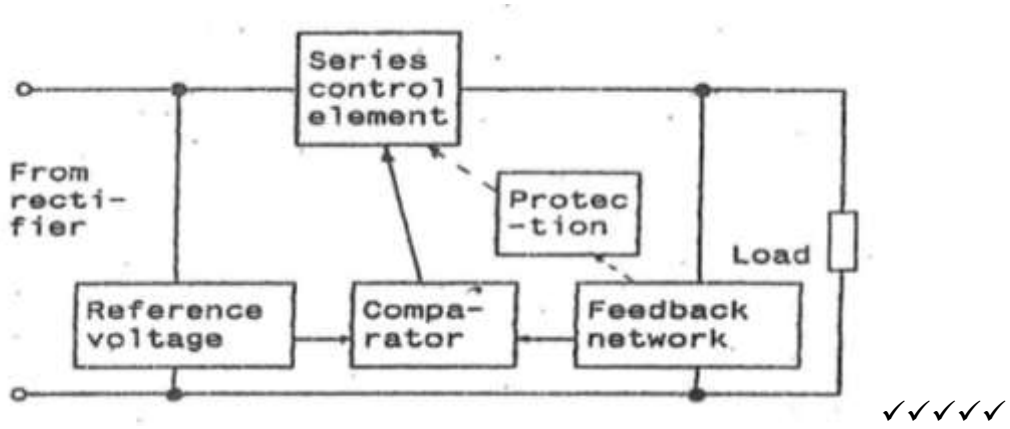
$$= 6{,}94\ V \checkmark$$

$$r' = \frac{V'_{r(rms)}}{V'_{DC}}$$

$$= \frac{6,94}{142,86} \checkmark$$

$$= 0,05 \checkmark \longrightarrow \quad (7)$$

2.2



✓✓✓✓✓

Block diagram of a series voltage regulator

1. Supplies reference voltage✓
2. Compares reference voltage and feedback✓
3. Supplies feedback voltage✓
4. Controls load current✓
5. Protects control element✓ (10)

[17]

QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 3.1.1 $I_C = \beta . I_B$

$$= 300 \times 26 \times 10^{-6} \checkmark$$

$$= 7,8 \text{ mA} \checkmark$$

$$R_C = \frac{V_{cc} - V_{ce}}{I_C}$$

$$\therefore V_{cc} = V_{ce} + R_C . I_C \checkmark$$

$$= 5 + 1,2 \times 10^3 \times 7,8 \times 10^{-3} \checkmark$$

$$= 14,36 \text{ V} \checkmark \longrightarrow \quad (5)$$

3.1.2 $R_b = \frac{V_{cc} - V_{be}}{I_b}$

$\therefore V_{be} = V_{cc} - R_b . I_b$ ✓

$= 14{,}36 - 525\,385 \times (26 \times 10^{-6})$ ✓

$= 0{,}7\ V$ ✓ ⟶ (3)

3.2 3.2.1 $Z_L = \frac{R_C \times R_L}{R_C + R_L}$

$= \frac{1\,200 \times 6\,000}{1\,200 + 6\,000}$ ✓

$= 1\,000\ \Omega$ ✓

$A_i = \left(\frac{h_{fe}}{1 + h_{oe} . Z_L}\right)$

$= \left(\frac{38}{1 + 6 \times 10^{-6} \times 1\,000}\right)$ ✓

$= 37{,}773$ ✓ ⟶ (4)

3.2.2 $A_V = \frac{-h_{fe} Z_L}{h_{ie} + (h_{ie} h_{oe} - h_{fe} h_{re})}$

$= \frac{-38 \times 1\,000}{1\,700 + (1\,700 \times 6 \times 10^{-6} - 38 \times 1{,}3 \times 10^{-6})}$ ✓✓

$= -22{,}353$ ✓ (3)

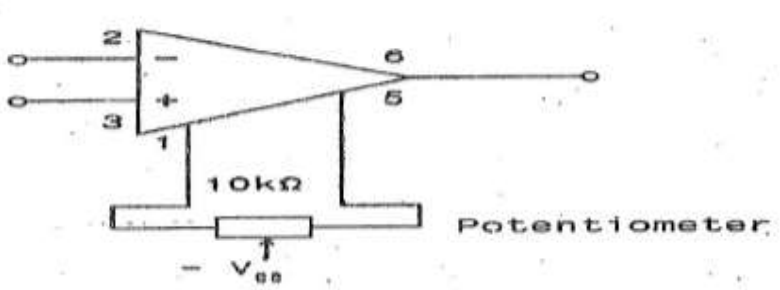
3.3 The amplified output signal of an amplifier ought to be similar to the input signal; therefore✓ if the output signal does not adopt the format of the input signal, the output signal is distorted.✓ (2)

[17]

QUESTION 4: OPERATIONAL AMPLIFIERS

4.1
- Very high input impedance that causes almost no current load for circuits
- Very high voltage gain suitable to amplify very small signals
- Very low output impedance not easily affected by other loads (3)

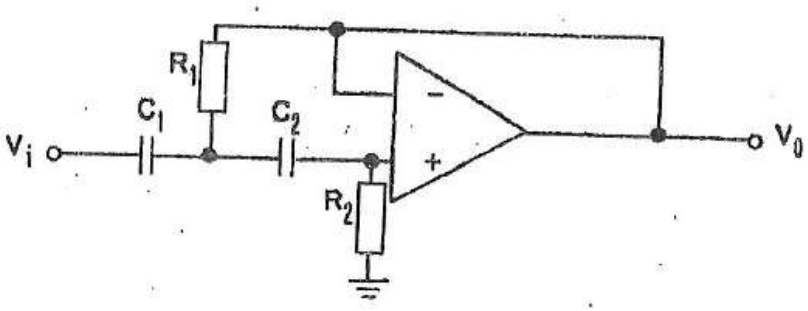
4.2



✓✓✓

Null-balance adjustment of a 741-operational amplifier (3)

4.3



✓✓✓

High-pass filter (3)

4.4 $f_o = \frac{1}{2\pi RC}$

$\therefore C = \frac{1}{2\pi R f_o}$

$= \frac{1}{2 \times \pi \times 10 \times 10^3 \times 150 \times 10^3}$ ✓

$= 10{,}61$ ρf ✓ ⟶ (2)

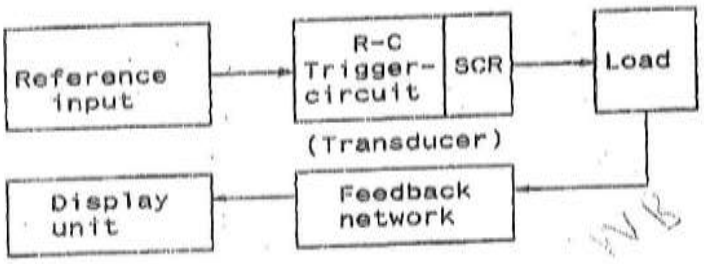
[11]

QUESTION 5: INTEGRATED CIRCUITS

5.1 integrated circuits
5.2 diodes
5.3 reverse
5.4 Zener diodes
5.5 15 V

(5 × 1) **[5]**

QUESTION 6: ELECTRONIC PHASE CONTROL



Open-loop system ✓✓✓✓✓ **[5]**

QUESTION 7: TEST EQUIPMENT

Presume that the unknown voltage V_u = 15 V and a-bit code is used:

Start: 0000✓

Step 1: 1000 (8 V) $V_r < V_u$: Sets 1✓

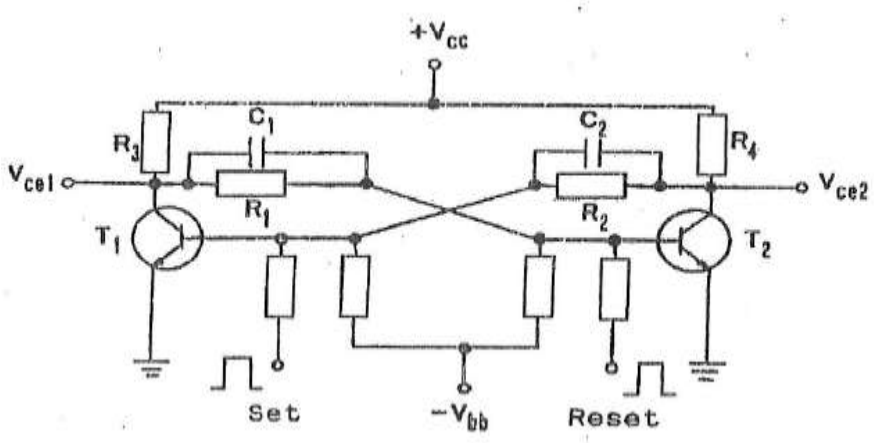
Step 2: 1011 (11 V) $V_r > V_u$: Reset 1 to 0✓

Step 3: 1010 (10 V) $V_r = V_u$: Sets 1✓

Step 4: 1100 (12 V) $V_r > V_u$: Reset 1 to 0✓ **[5]**

QUESTION 8: OSCILLATORS

8.1



✓✓✓✓

Bi-stable multivibrator (4)

8.2 $f_O = \frac{1,5}{RC}$

$\therefore C = \frac{1}{Rf_O}$

$= \frac{1,5}{15 \times 10^3 \times 10}$ ✓

$= 10\ \mu f$ ✓ ⟶ (2)

8.3
8.3.1 False
8.3.2 False
8.3.3 False
8.3.4 True
8.3.5 True
8.3.6 True

(6 × 1) (6)
[12]

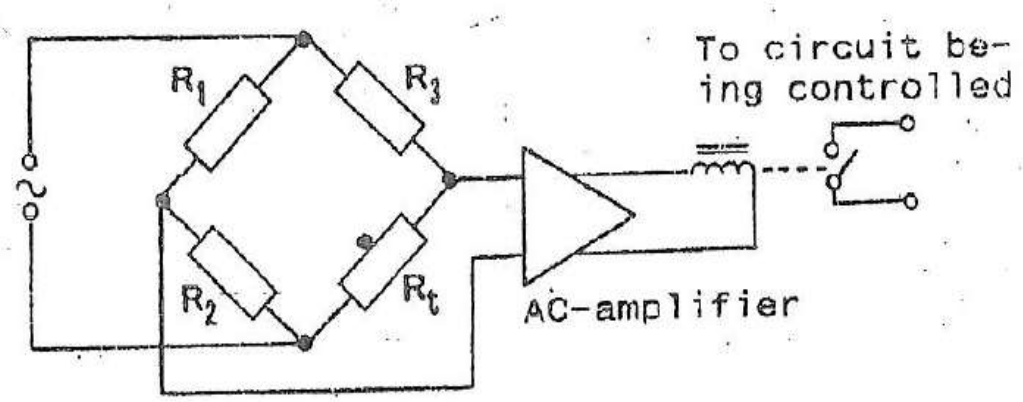
QUESTION 9: TRANSDUCERS

9.1 The resolution of a potentiometer can be defined as the relation of the voltage drop across two adjacent turns to the total voltage drop or the relation of one turn to the total number of turns of which the resistor consists. (2)

9.2 The slider of the feedback potentiometer can be adjusted mechanically by the axle of the canon turret and sends a feedback signal to the comparator. The comparator compares the feedback signal with the reference signal which is adjusted manually. If there is a difference, the error signal is given and applied to the motor. The motor rotates the gearbox and canon respectively, which rotates the axle of the feedback potentiometer in return. As the canon nears its required position, the error signal becomes smaller. As soon as the canon reaches its required position, the error signal is zero and the motor will stop. (5)

9.3 Resistive transducers use the change in resistance to convert other forms of energy✓ like displacement, heat, light, power, torsion, humidity, etc. to proportional current or voltage.✓ (2)

9.4



Temperature control with the aid of a thermistor (4)
[13]

TOTAL: 100

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